

Multiplication and division

Multiplication and division help us work efficiently with equal groups. In Year 3, pupils build rapid recall of the 3, 4 and 8 times tables while developing a clear understanding of how the two operations are connected.

One fact family - four connected facts

If pupils know one multiplication fact, they can use the inverse relationship to derive division facts.

$$4 \times 8 = 32$$

$$8 \times 4 = 32$$

$$32 \div 4 = 8$$

$$32 \div 8 = 4$$

Times-table connections

Doubling the 2 times table produces the 4 times table. Doubling again produces the 8 times table. This reveals a useful pattern instead of three isolated sets of facts.

START

$$2 \times 6 = 12$$

**DOUBLE**

$$4 \times 6 = 24$$

**DOUBLE**

$$8 \times 6 = 48$$

Everyday life examples

PACKS

$$4 \text{ packs} \times 8 \text{ biscuits}$$

32 biscuits altogether

SHARING

$$12 \text{ sweets} \div 3 \text{ children}$$

4 sweets each

ARRAYS

$$8 \text{ rows} \times 4 \text{ chairs}$$

32 chairs altogether

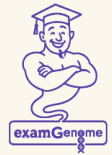
OUTFITS

$$3 \text{ hats} \times 4 \text{ coats}$$

12 combinations

WHY FLUENCY MATTERS

Fast recall frees pupils to focus on reasoning, choosing operations and solving problems.



Using multiplication and division

Scaling and correspondence

Multiplication describes how quantities scale and how choices combine. Division can share a quantity equally or determine how many equal groups can be made.

RECIPE SCALE

2 → 8 people: $\times 4$

Multiply every quantity by 4

MAP SCALE

$5 \times 3 \text{ km} = 15 \text{ km}$

At 3 km for every 1 cm

COMBINATIONS

3 hats $\times$ 4 coats

12 possible outfits

Efficient mental methods

Numbers can be rearranged in multiplication to create easier steps. The value stays the same, but the calculation becomes quicker.

$$4 \times 12 \times 5 = 4 \times 5 \times 12 = 20 \times 12 = 240$$

Moving towards written methods

For two-digit by one-digit calculations, partitioning makes each place-value step visible and prepares pupils for short multiplication and division.

MULTIPLICATION

$23 \times 4 = 92$

$(20 \times 4) + (3 \times 4) = 80 + 12$

DIVISION

$84 \div 4 = 21$

$(80 \div 4) + (4 \div 4) = 20 + 1$

By the end of this topic, pupils should be able to:

- recall facts for the 3, 4 and 8 times tables
- use related multiplication and division facts
- calculate two-digit numbers by one-digit numbers
- solve sharing, grouping and scaling problems
- solve correspondence and combination problems
- explain and check an efficient chosen method

FOUNDATION FOR THE FUTURE

Secure multiplication and division support fractions, area, percentages, scaling and algebra.